

KUAN BUTTS

San Francisco Bay Area | Staff Software Engineer
Distributed Systems, Data Infrastructure, Geospatial

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SUMMARY

Staff-level software engineer specializing in distributed data systems, real-time telemetry pipelines, and large-scale data processing. 10+ years experience designing and building high-throughput backend systems that operate at global scale. Particularly strong in systems design, data ingestion pipelines, event streams, and model-driven inference in production.

EXPERIENCE

Mapbox — Staff Software Engineer

San Francisco, CA 2019–Present

Built and operated large-scale distributed systems powering global real-time traffic and telemetry products used by enterprise customers.

Select accomplishments:

- Designed and deployed a **distributed inference pipeline processing 5B+ location observations/hour** for live speed estimation.
- Re-architected the global speed calculation system into a decoupled, high-accuracy inference service, yielding up to **60%** surface-street coverage improvements.
- Built ML-driven data pipelines for modality classification and anomaly detection using raw telemetry and sensor data.
- Delivered a **global, daily-updated activity index** system with segment- and block-level granularity, enabling analytics for enterprise and mobility customers.
- Optimized data ingestion, matching, and inference workflows for reliability, latency, and cost efficiency.

Tech: Python, TypeScript, AWS/GCP, Kafka/Kinesis, Spark, Airflow, Redis/ValKey, Docker

UrbanFootprint — Data Science & Engineering

Berkeley, CA 2017–2019

Full-stack data engineer building data processing pipelines and compute workflows behind a large-scale analytics platform.

- Developed high-performance graph algorithms and routing models.
- Built ingestion workflows for large telemetry datasets and optimized long-running batch jobs.
- Delivered large data integrations and analytics products for enterprise customers.

Tech: Python, TypeScript, Postgres/PostGIS

Code for America — Software Engineer

San Francisco, CA 2016

- Converted a single-tenant web platform (ClientComm) into a **scalable multi-tenant SaaS application** deployed by multiple jurisdictions.
- Implemented data ingestion & analytics for text-based communications at scale.

Tech: Python, Flask, Node, Postgres, Redis, Heroku/Azure

Microsoft — Civic Technology Fellow

New York, NY 2015

- Built data ingestion pipelines for real-time NYC MTA vehicle data and enabled analytics on transit performance.
- Designed early prototypes for real-time signal monitoring workflows.

MIT / MIT Media Lab — Software Developer & Research Engineer

Cambridge, MA 2012–2015

- Developed mobile and web-based data collection tools and geospatial data processing pipelines.
- Built interactive analytics tools for transportation researchers; collaborated across multidisciplinary engineering teams.

EDUCATION

Massachusetts Institute of Technology (MIT) - MCP (Master's)

Master of City Planning, Transportation Systems Planning *2012–2014*

Thesis: **Iterative Methods for Large-Scale Mobile Data Acquisition**